Does the Hicks $T(3,4)$ vortex line support a finite stable vortex tube?

A063 v0.2.0 — analytical/numerical reference report

Outcome: The closed $T(3,4)$ vortex-field-line is independently classified numerically. The tested constant-radius knotted tubular surface is **not material**, and neighboring winding shows nonzero shear. The existence and nonlinear stability of a differently shaped finite tube remains **UNRESOLVED**.

7 PASS / 2 FAIL / 3 UNRESOLVED

1. Model and falsifiable distinctions

Inside the spherical Hicks aggregate ($r < a$), the exact axisymmetric streamfunction is

$$\psi = A\big[J(\lambda r/a) - (r/a)^2 J(\lambda)\big]\sin^2\theta, \quad J(y) = \frac{\sin y}{y} - \cos y. \tag{1}$$

With (s, ϕ, z), $u_s = -\partial_z\psi/s$, $u_z = \partial_s\psi/s$, $u_\phi = (\lambda/a)\psi/s$; the first root of $\tan\lambda = \lambda$ is $\lambda_1 \simeq 4.4934094579$. In this Beltrami branch, $\boldsymbol{\omega} = (\lambda/a)\mathbf{u}$, hence a streamline is also a *vorticity line*.

Separate four propositions: (I) a closed knotted line; (II) an embedded geometric neighborhood; (III) an actual vortex-aligned/material surface $\boldsymbol{\omega}\cdot\mathbf{n} = 0$; (IV) nonlinear time-dependent Euler stability. (I) never implies (III) or (IV).

2. Reproduced orbit and independent topology test

A new high-accuracy integration gives $s_0/a = \mathbf{0.968309830174}$, winding $W = \mathbf{0.75000000000010}$ and closed-orbit residual $\|X(T) - X(0)\|/a = \mathbf{8.7311e-10}$ over four poloidal turns.

The 3-D polygon was projected along four distinct generic directions (raw crossing counts: 11 / 9 / 13 / 15). For each projection, over/under crossing order yields a Wirtinger/Fox Alexander matrix, and its minor gives the same normalized polynomial:

$$\Delta_{\text{numerical}}(t) = t^6 - t^5 + t^3 - t + 1 = \Delta_{T(3,4)}(t), \qquad |\Delta(-1)| = 3. \tag{2}$$

This is independent of a mere winding-label assignment. Matching the Alexander polynomial is not a unique-knot theorem in isolation; the numerically embedded invariant torus and $(3,4)$ winding supply additional classification evidence.

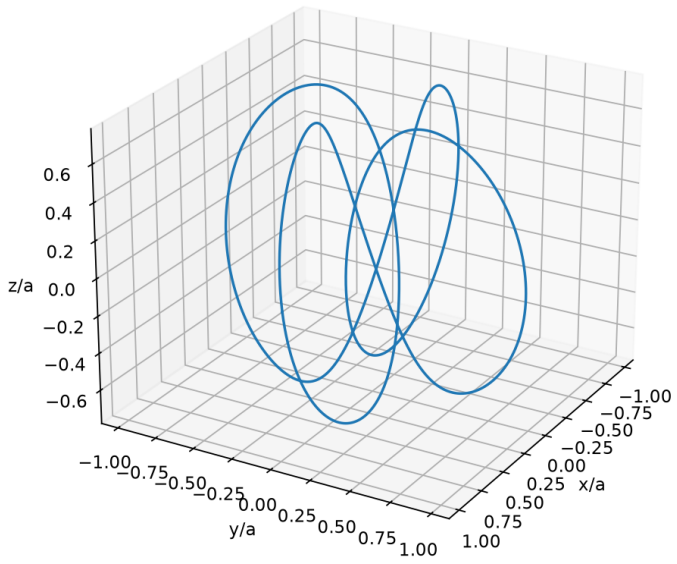


Figure 1: Numerical $T(3,4)$ Hicks-Beltrami vortex line in dimensionless Cartesian coordinates.

3. Geometric thickening versus material vortex tube

The sampled minimum curvature-radius proxy is $R_\kappa/a \simeq 0.41864$ and the excluded-neighbor point-distance proxy is $d_{\mathrm{nonlocal}}/a \simeq 0.20135$. The orbit stays at least $0.031690a$ inside the spherical separatrix. These are **not continuum tube reach bounds** and cannot certify absence of every self-intersection for an arbitrary surface.

For the explicit normal tube $X_\delta(\sigma,\vartheta) = X(\sigma) + \delta[\mathbf{N}(\sigma)\cos\vartheta + \mathbf{B}(\sigma)\sin\vartheta]$, a material steady tube must satisfy $\mathbf{u}\cdot\mathbf{n} = 0$ pointwise; and the Beltrami vortex flux satisfies $\boldsymbol{\omega}\cdot\mathbf{n} = (\lambda/a)\mathbf{u}\cdot\mathbf{n}$. We sampled the actual velocity using the exact field (90 longitudinal points, 16 angular samples, plus a finer 180×24 check).

δ/a	RMS $\lvert u_n\rvert/u_{\mathrm{rms}}$	max $\lvert u_n\rvert/u_{\mathrm{rms}}$
0.0020	0.0024601	0.0039760
0.0040	0.0049202	0.0079704
0.0080	0.0098400	0.0160367

Table 1: Pointwise normal-velocity diagnostic for an initially constant-radius knot tube.

The zero-radius tangent control is 3.854e-17. The normal leakage is substantially above the preregistered 10^{-4} threshold; therefore **this steady constant-radius normal tube fails the material-surface gate (T5)**. This does not rule out a nonconstant-radius or advected time-dependent tube.

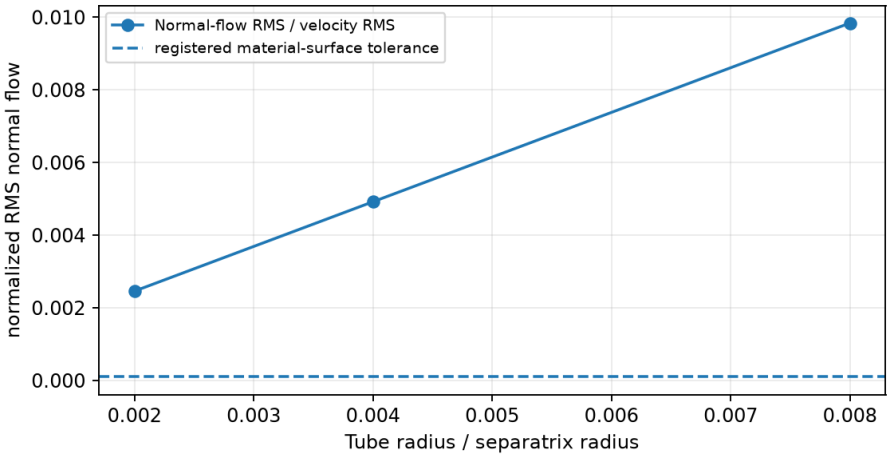


Figure 2: Measured positive normal flow increases with tube radius.

4. Nearby field lines and pitch-shear test

The Poincaré return rotation $W(s_0)$ has derivative $dW/d(s_0/a) \simeq -0.746819$. Thus a nonzero radial displacement develops azimuthal phase slip after four poloidal circuits:

$$\Delta\phi_4 \simeq 8\pi\,[W(s_0+\delta s) - 3/4]. \tag{3}$$

Therefore a positive-width annulus of neighboring lines does not share the exact same 3:4 closure under this integrable Hicks flow. **T6 FAIL is a field-line-coherence result, not a time-dependent Euler instability eigenvalue.**

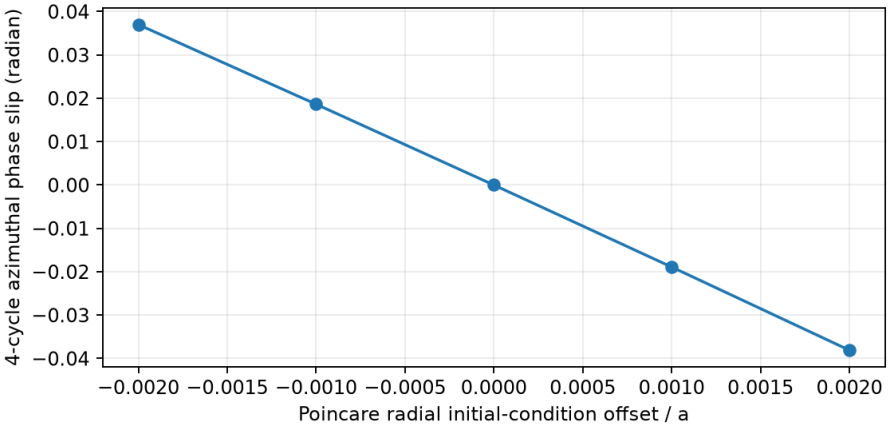


Figure 3: Neighboring initial conditions produce opposite signed four-period slip.

5. Finite-time material advection in the prescribed Hicks field

A material knotted surface (initial $\delta/a = 0.004$, 384 markers; confirmation with 720 markers) was transported with $\dot{X} = u_{\mathrm{Hicks}}(X)$ while the steady Euler field itself remained fixed. The surface's RMS distance to the original closed knot after four core periods rose to 12.729 times its initial radius (confirmation sample: 12.955 times); intermediate records are stored in the JSON output. This is finite-time kinematic deformation, not a coupled perturbed 3-D Euler evolution.

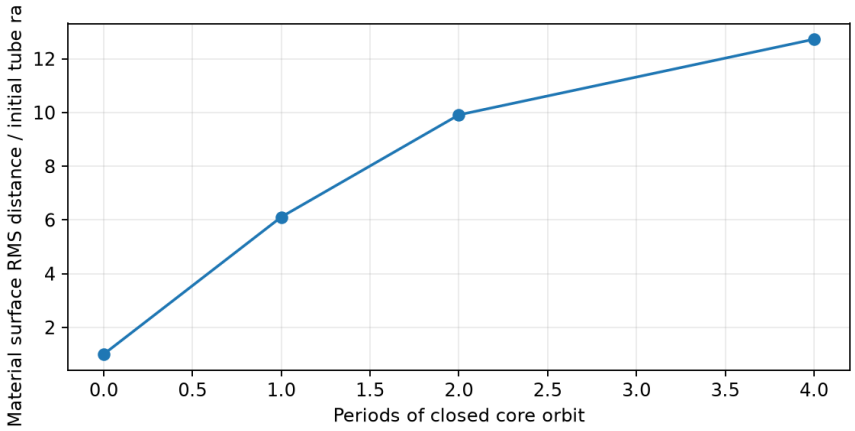


Figure 4: Localization of the advected material surface relative to the closed reference knot.

6. Gate ledger and continuation policy

Gate	Result	Qualification question
`T0`	PASS	Release source/protocol hashes recorded
`T1`	PASS	Reproduce a closed rational 3:4 Hicks vortex line
`T2`	PASS	Independent Alexander polynomial across generic projection directions
`T3`	PASS	Sampled geometric regularity and confinement of a finite normal tube
`T4`	PASS	The core is a vortex line: omega=(lambda/a)u
`T5`	FAIL	Is a constant-radius knotted tube an instantaneously material/vorticity-aligned surface?
`T6`	FAIL	Do adjacent vortex lines have the same 3:4 return rotation (no pitch shear)?
`T7`	PASS	Distributed full-sphere vorticity control (not a compact isolated knot tube)
`T8A`	PASS	Prescribed-field material advection deformation documented
`T8`	UNRESOLVED	True finite-core stability under 3D perturbed Euler evolution
`T9`	UNRESOLVED	Independent C++ FP64 and canonical v1.0.7 FREEZE/CERTIFY

T10	UNRESOLVED	Can any finite topologically consistent dynamically stable vortex tube be constructed?

A scientific FAIL never stops later diagnostic gates. T8A PASS certifies execution of the prescribed-field deformation measurement, not stability. Gates T8 and T9 remain open, rather than being relabeled PASS: no three-dimensional, pressure-coupled perturbed Euler evolution or independent native C++ FP64/canonical-framework certification has been performed.

7. What a decisive follow-up must demonstrate

Pre-register: a finite smooth tubular surface with independent knot invariant and positive clearance; a flux condition $\boldsymbol{\omega} \cdot \mathbf{n} = 0$ (or an explicitly time-dependent material-boundary evolution law); constrained-energy stationarity; reproducible linearized-Euler spectra; three spatial resolutions, at least three perturbation amplitudes, and nonlinear bounded deformation for 10 core periods, with energy/helicity, divergence, topology and pressure-Poisson audits. Source-independent carriers must be identified separately (E013), not silently substituted.

Strict conclusion. The observed $T(3, 4)$ is a convincing *numerical vortex-line knot*, and a finite geometric neighborhood can be sampled. A naive steady constant-radius vortex tube is rejected. No physical particle, persistent finite vortex core, full 3D stability or mass quantization follows.

References and provenance

1. W. M. Hicks (1899), *Researches in Vortex Motion. Part III: On Spiral or Gyrostatic Vortex Aggregates*, doi:10.1098/rsta.1899.0002.

2. M. J. M. Hill (1894), *On a Spherical Vortex*, doi:10.1098/rsta.1894.0006.

3. A063-v0.1.1: numerical streamline precursor. A063-v0.2.0: release source seal, gate ledger, controls JSON and vortex-line NPZ. All evidence is numerical, not experimental.